Unified chassis control: a model-free approach

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Abstract

Automotive safety has been a critical focus for both national and international research institutions in recent years. The combined control of lateral dynamics and roll stability in vehicles has been extensively studied, often relying on automotive models to design and tune controllers. However, these models frequently include parametric uncertainties, nonlinear dynamics, and simplifying assumptions, which can lead to suboptimal controller performance in real-world applications. This work proposes a unified chassis controller (UCC) to regulate lateral and roll dynamics during abrupt maneuvers, comparing two control strategies: one based on traditional PID controllers and another based on model-free controllers (MFC). Both approaches employ three independent controllers to maintain the side-slip angle, yaw rate, and roll angle within safe limits. The robustness and performance of these control structures is evaluated using a Sine With Dwell maneuver in a co-simulation environment combining MATLAB/Simulink® and CarSim®. Results demonstrate that while the PID-based UCC struggles to maintain stability under varying road conditions, the MFC-based UCC achieves superior robustness and performance, effectively regulating the state variables across different scenarios. These findings highlight the robustness and adaptability of MFCs, making them a promising solution for enhancing vehicle safety.

Keywords: UCC, robustness, CarSim, MFC

1 Introduction

Road traffic accidents remain a major global concern, causing approximately 1.3 million deaths and tens of millions of injuries each year (Naciones Unidas (Noticias ONU), 2022). Many of these incidents are associated with loss of vehicle stability during abrupt maneuvers, tire failures, or adverse road conditions (Friedman et al., 2010). Such situations are typically characterized by excessive side-slip angle and yaw rate, which can lead to oversteer or understeer, as well as increased rollover risk due to significant roll angle variations (Stolle et al., 2011; Zhan, 2017; Nguyen, 2023). Consequently, side-slip angle, yaw rate, and roll angle have been widely recognized as the key state

variables for vehicle stability monitoring and control, serving as the primary feedback signals in active safety systems (Rajamani, 2012).

To address these challenges, modern vehicles incorporate advanced control systems such as Electronic Stability Control (ESC), which improves lateral stability through selective braking and torque modulation (Aleksander and Edward, 2007). In parallel, suspension-based systems, including active and semi-active suspensions, are used to regulate vertical dynamics and enhance ride comfort.

Unified Chassis Control (UCC) extends these concepts by coordinating multiple subsystems to simultaneously regulate lateral and vertical vehicle dynamics, improving overall stability and safety, particularly under extreme driving conditions (Wanki et al., 2012). However, achieving this coordination is non-trivial, as lateral stability enhancement, rollover mitigation, and actuator allocation are inherently competing objectives, requiring careful trade-off management to avoid control conflicts between subsystems (Falcone et al., 2007).

Most existing UCC strategies rely on model-based control approaches, which can be broadly categorized by their underlying vehicle representation. On one end, simplified formulations such as the bicycle model have been widely adopted within Model Predictive Control (MPC) frameworks, offering computational tractability for real-time path-following and yaw stabilization tasks (Ahmadi, 2021; Gidlewski et al., 2017; Yakub et al., 2016). On the other end, multi-degree-of-freedom formulations have been proposed to capture the coupling between lateral, vertical, and suspension dynamics more faithfully: Lai and Jiang (2021) developed an 18-DOF chassis model that enables high-fidelity simulation of cross-subsystem interactions, while Chen et al. (2021) extended this perspective to electric vehicles by explicitly incorporating wheel vertical vibrations into a UCC architecture.

Although these high-fidelity representations improve prediction accuracy and allow for a more principled treatment of lateral-vertical coupling, they introduce substantial computational overhead and exhibit strong sensitivity to parametric uncertainty, limiting their portability across varying road conditions and vehicle configurations (Guo et al., 2020; Lew et al., 2024).

To mitigate these limitations, several works have introduced uncertainty compensation mechanisms based on data-driven or adaptive techniques. Neural network-based controllers have been widely used to approximate unknown nonlinearities and disturbances in vehicle dynamics, enhancing robustness under varying conditions. Similarly, fuzzy and neuro-fuzzy approaches enable online adaptation of control parameters, while sliding mode control combined with intelligent approximators provides robustness against disturbances and model mismatch (Swain et al., 2021; Guo et al., 2021).

More recently, hybrid strategies combining model-based controllers with learning-based compensation terms have demonstrated improved performance in the presence of uncertainties (Levine et al., 2016). While these strategies collectively advance robustness against parametric uncertainty and external disturbances, a systematic and homogeneous comparison between them and model-free control approaches within a UCC architecture remains scarce in the literature, limiting the ability to draw generalizable conclusions about their relative merits under consistent evaluation conditions (Sánchez-Morales et al., 2023; Mazzilli et al., 2021).

Despite the advances described above, the practical deployment of UCC strategies under varying operating conditions remains challenging. Model-based approaches, even when augmented with adaptive or learning-based compensation mechanisms, often remain sensitive to parametric uncertainty and typically require significant retuning when vehicle parameters or road conditions change.

In this context, model-free control (MFC) has emerged as a structurally simpler alternative that reduces dependence on an explicit vehicle model; however, its performance within a UCC architecture has only rarely been assessed against representative baselines. In particular, many comparative studies in this domain rely primarily on fixed-gain PID controllers, leaving open the question of how MFC compares with more meaningful references such as gain-scheduled controllers, simulator-embedded baseline strategies,

and learning-based alternatives under consistent evaluation conditions.

To address this gap, this paper proposes a model-free UCC strategy for the simultaneous regulation of side-slip angle, yaw rate, and roll angle, and evaluates it in a MATLAB/Simulink-CarSim co-simulation environment under a sine-with-dwell maneuver with different road friction coefficients. The main contributions of this work are threefold: first, the design and implementation of an MFC-based UCC architecture for the joint regulation of lateral and vertical vehicle dynamics without relying on an explicit vehicle model; second, a comparative assessment against representative benchmark controllers, including a fixed-gain PID, a gain-scheduled PID, and a CarSim built-in baseline; and third, an analysis of practical closed-loop stability, robustness to friction variation, and control performance limits of the proposed approach.

The remainder of the paper is organized as follows. Section 2.4 presents the modeling framework and reviews model-based UCC approaches. Section 3 introduces the fundamentals of MFC and the design of intelligent PID (iPID) controllers. Section 4 describes the proposed UCC architecture. Section 5 discusses controller tuning, and Section 6 presents the experimental setup. Finally, Section 7 discusses the results, and Section 8 concludes the paper.

2 Control Problem Formulation

2.1 Vehicle States and Control Objectives

2.2 Reference Generation and Operating Constraints

2.3 Maneuver Completion and Failure Criteria

2.4 UCC model-based

Currently, most of the UCC systems use vehicle models to obtain and adjust the parameters of the individual controllers.

A widely adopted approach is the bicycle model, illustrated in Fig. 1. In this representation, F_f and F_r denote the lateral forces acting on the front and rear axles, respectively; V_{ref} is the vehicle's velocity vector; and α_f and α_r are the slip angles of the front and rear wheels. The Center of Gravity (CG) is located at distances l_f and l_r from the front and rear axles. The side-slip angle, β, represents the relationship between the vehicle's longitudinal and lateral velocities, while $r = \dot{\psi}$ denotes the yaw rate. Additionally, δ_f epresents the front wheel steering angle, and V_f and V_r are the velocity vectors at the front and rear tires.

This model is frequently integrated into Model Predictive Control (MPC) frameworks, as seen in Ahmadi (2021), Gidlewski et al. (2017), and Yakub et al. (2016). Other studies, such as Owen (2018), utilize it to analyze the impact of the center-of-mass location on vehicle performance, while Bilen (2010) and Fahimi (2013) employ it for computational simulations and controller design (Xia et al., 2021).

To achieve higher fidelity, some authors develop more complex models to capture intricate vehicle dynamics and subsystem interactions. For instance, Lai and Jiang (2021) introduced an 18-degree-of-freedom (DOF) model to represent comprehensive chassis dynamics. This model facilitates the design of a UCC system that evaluates the trade-off between ride comfort and handling stability across various maneuvers, accounting for the complex coupling between subsystems. The 18-DOF model treats the vehicle as five rigid bodies: a sprung mass (chassis) with 6-DOF (roll, pitch, yaw, and translational motions) and four unsprung masses (wheels). Each wheel is modeled with 3-DOF, corresponding to vertical displacement, wheel rotation, and steering rotation.

Similarly, Chen et al. (2021) proposed a 6-DOF rigid-body model that integrates planar motion (longitudinal, lateral, and yaw) with spatial motion (vertical, roll, and pitch). Combined with unsprung mass dynamics, this approach treats the vehicle as a unified system under longitudinal and lateral accelerations. To characterize nonlinear tire behavior, the authors utilize Pacejka's Magic Formula, an empirical approach that unifies longitudinal force, lateral force, and self-aligning moment through parameter sets fitted to experimental data.

The use of model-based controls in vehicle dynamics offers the fundamental advantage of enabling precise and coordinated management of complex multivariable systems—such as the

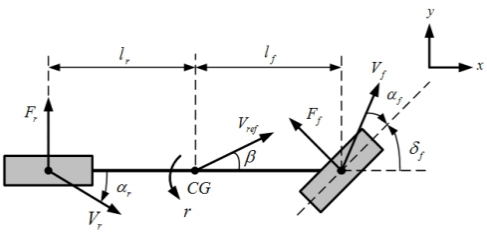


Fig. 1: Bicycle model

interaction between independent wheel drive and active suspension—facilitating the incorporation of physical constraints and the prediction of critical states to maximize safety and comfort Li et al. (2022). However, this approach has critical drawbacks, primarily a high demand for computational resources to process high-fidelity models in real time Guo et al. (2020), as well as significant sensitivity to modeling errors (model mismatch) Lew et al. (2024). Any discrepancy between mathematical parameters and real-world conditions—such as variations in mass or unexpected changes in tire friction—can significantly degrade system performance and stability if robust adaptation mechanisms are not in place.

2.5 Uncertainty Compensation

Given the limitations discussed above, neural network-based controllers have been widely used to compensate for model uncertainties and unmodeled nonlinearities in vehicle dynamics. This is the case in Swain et al. (2021), where the lateral dynamics are described using a dynamic bicycle model, capturing key variables such as longitudinal velocity, slip angle, yaw rate, and tire lateral forces, while also accounting for trajectory tracking errors and unmodeled external disturbances (e.g., crosswinds). To compensate for parametric uncertainties, nonlinearities, and unmodeled dynamics, a radial basis function neural network (RBFNN) is integrated into a higher-order sliding mode control (HOSMC) scheme. The RBFNN estimates the unknown dynamics online based on the sliding surface, with its parameters updated through Lyapunov-based adaptive laws. This combined approach enhances robustness against disturbances, reduces chattering, and preserves lateral control performance under adverse conditions.

In addition to neural network-based approaches, fuzzy logic has also been employed to handle parametric uncertainties within sliding mode control frameworks. This is the case in Guo et al. (2021), where a fuzzy inference system is used to adaptively tune the boundary layer thickness based on the magnitude of the sliding surface, as part of a neuro-fuzzy adaptive sliding mode control strategy.

In this approach, the absolute value of the sliding surface is used as the input to the fuzzy system, while the boundary layer thickness is the output. Using predefined linguistic rules and membership functions, the controller adjusts this parameter online: large deviations from the sliding surface lead to a thinner boundary layer to ensure fast convergence, whereas small deviations result in a thicker layer to attenuate chattering.

This adaptive fuzzy mechanism enables an effective trade-off between robustness and oscillation reduction, improving control performance in the presence of parametric uncertainties and external disturbances.

One of the most widely used schemes for validating controller design in automotive systems is the CarSim® software. CarSim® mathematical models are highly optimized for computational efficiency and are represented at the system level, meaning that vehicle data is intended to be measured or calculated and does not rely on detailed knowledge of component materials or suspension linkage complexities. When using a CarSim® model, major issues may include the need for detailed vehicle data, accurate environmental conditions, and a good understanding of vehicle models at the system level. Users should also be aware of the importance of accurate kinematics and suspension compliance, tire strength and moment properties, and environmental conditions, as well as the potential impact of limited reference data on the accuracy of simulation results (Mechanical Simulation Corporation, 2021).

Due to the above, the study of MFC for the regulation of vehicle dynamics in real-time is of interest. It should be noted that the term "Model-free control" can be seen in the literature as referring to a wide range of controllers that can go from "classical" proportional-integral-derivative (PID) to robust and adaptive control techniques derived, for example, from neural networks, fuzzy systems, and soft computing. The type of MFC used in

this paper (intelligent PID based on ultra-local model) was presented by Fliess and Join (2008), which makes it a novel technique and alternative to classical control in the automotive domain. Several authors have used this control technique as part of an automotive UCC system. In general, the MFC can be used in this field to regulate the longitudinal dynamics (direction and/or speed) of the vehicle (Dong et al., 2014), as a controller of the lateral dynamics (yaw rate and/or side slip angle) as in the case of Radac and Precup (2018), for the control of the joint lateral and longitudinal dynamics like in Menhour et al. (2018) and Rizk et al. (2023), or for the control of the vertical dynamics, that is roll, roll rate and/or pitch (Mustafa et al., 2019).

However, there are no references in the literature to works that use this control method to jointly regulate the lateral and vertical dynamics at the same time and in a single UCC system. In the present work, an automotive UCC system is designed using only MFC controllers to minimize the side-slip angle, reducing the yaw rate error and maintaining the vertical stability corresponding to the car roll angle and roll rate. The proposed system is tuned and validated through a Sine With Dwell maneuver in CarSim® software. Those are original contributions, to the best of the authors' knowledge.

3 The ultra-local model-based MFC controller

The Model-Free Controller (MFC) based ultra-local model assumes that the system response will behave as shown in Eq. (1) for a small time window:

$$y^v = F + \alpha u, \tag{1}$$

where y^v represents the vth derivative of the output (usually $v = 1$ or $v = 2$) (Fliess, 2009), and F is the input/output ratio of a simplified and little-known model of the plant. This parameter is a piecewise constant function that is updated in each time window and also includes measured and unmeasured perturbations of the plant. On the other hand, $\alpha \in \mathbb{R}$ is a number that represents a non-physical constant parameter. This parameter is defined by the designer in such a way as to have the same order of magnitude as the unknown real parameter, and then an adjustment is made based on the system's time response. In this paper, there are three plant variables: yaw rate ($\dot{\psi}$), side-slip angle (β), and roll angle (ϕ), where the last two ones are controlled using a second-order ultra-local model, while the first one is represented by a first-order ultra-local model.

Assuming that the model represents the plant dynamics in a short time window, then the stabilizing intelligent PID control (iPID) can be computed by:

$$u = \frac{-F}{\alpha} + \frac{\ddot{y}^*}{\alpha} + K_P e + K_I \int e\, dt + K_D \frac{de}{dt}, \tag{2}$$

where $\ddot{y}^*$ is the reference second time-derivative that the output is intended to follow, $e = y - y^*$ is the system tracking error and K_P, K_I and K_D are the gains of a conventional PID controller. These gains are adjusted to cancel out the measured output and impose the desired dynamics. It should be noted that this control can be in the form of a PI or a PD as appropriate. Finally, the control method involves estimating the value of F in constant time windows. It can be demonstrated that even if the initial guess for α is incorrect, the estimator will correctly estimate the compensation of F (Pereira das Neves and Augusto Angélico, 2022). Therefore, α will only slightly affect the performance but will ensure the system stability.

3.1 F estimator to the ultra-local model of β and ϕ

As mentioned earlier, the ultra-local models of β and ϕ are considered of second order ($v = 2$). In that case, the equation would be $\ddot{y} = F_2 + \alpha u$. Applying the Laplace transform in this equation results in $s^2Y(s) - sy(0) - \dot{y}(0) = F_2/s + \alpha U(s)$. Then, by taking the derivative of the equation concerning s twice, one obtains:

$$2Y(s) + 4s\frac{d}{ds}Y(s) + s^2\frac{d^2}{ds^2}Y(s) = \\ = 2s^{-3}F_2 + \alpha\frac{d^2}{ds^2}U(s). \tag{3}$$

Finally, multiplying Eq. (3) by $\frac{1}{s^3}$ results in[1]:

$$2\frac{1}{s^3}Y(s) + 4\frac{1}{s^2}\frac{d}{ds}Y(s) + \frac{1}{s}\frac{d^2}{ds^2}Y(s) = \\ = 2F_2\frac{1}{s^6} + \alpha\frac{1}{s^3}\frac{d^2}{ds^2}U(s). \quad (4)$$

Returning to the time domain using the Laplace anti-transform, it is obtained:

$$F_2 = \frac{60}{t^5}\int_0^t (t^2 - 6t\tau + 6\tau^2)y(\tau)\,d\tau + \\ - \frac{30\alpha}{t^5}\int_0^t (t-\tau)^2\tau^2 u(\tau)\,d\tau. \quad (5)$$

3.2 F estimator to the ultra-local model of $\dot{\psi}$

The ultra-local model of $\dot{\psi}$ is considered of first order ($v = 1$). In that case, the equation would look like $\dot{y} = F_1 + \alpha u$. The demonstration is very similar to the estimator of second order. Applying the Laplace transform in this equation results in $sY(s) - y(0) = F_1/s + \alpha U(s)$. Then, by taking the derivative of the equation by s, one has:

$$Y(s) + s\frac{d}{ds}Y(s) = -s^{-2}F_1 + \alpha\frac{d}{ds}U(s). \quad (6)$$

Finally, multiplying the Eq. (6) by $\frac{1}{s^2}$ results in:

$$\frac{Y(s)}{s^2} + \frac{1}{s}\frac{d}{ds}Y(s) = -F_1 + \frac{\alpha}{s^2}\frac{d}{ds}U(s). \quad (7)$$

Upon returning to the time domain via the Laplace inverse transform, one obtains:

$$F_1 = -\frac{6}{t^3}\int_0^t (t - 2t\tau)y(\tau)\,d\tau + \\ - \frac{6\alpha}{t^3}\int_0^t (t-\tau)\tau u(\tau)\,d\tau. \quad (8)$$

3.3 Discretization of algebraic estimators

The computational application of the estimator is possible through the window integral approximation. For this, the size of the integration window (T) must be decided, where, in general, the larger the window, the more precise the estimated value will be. On the other hand, the control will be slower since the estimator will take longer to converge to the value. Considering the response speed of the car dynamics, the windows for all estimators were defined as $T = 1.0$ s.

Considering the trapezoidal approximation of the integral, the estimators (of first and second order) can be written as:

$$\hat{F}_2 = \frac{60}{T^5}\sum_{k=0}^{T/T_s} a_k(T^2 - 6TkT_s + 6(kT_s)^2)y[k] + \\ - \frac{30\hat{\alpha}}{T^5}\sum_{k=0}^{T/T_s} a_k(T - kT_s)^2(kT_s)^2 u[k], \quad (9)$$

and

$$\hat{F}_1 = -\frac{6}{T^3}\sum_{k=0}^{T/T_s} a_k(T - 2TkT_s)y[k] + \\ - \frac{6\hat{\alpha}}{T^3}\sum_{k=0}^{T/T_s} a_k(T - kT_s)kT_s u[k], \quad (10)$$

where:

$$a_k = \begin{cases} T_2/2, & k = 0 \text{ and } k = T/T_s \\ T_s, & k = 1, \ldots, T/T_s - 1 \end{cases}. \quad (11)$$

4 UCC structure

In this work, an upstream architecture with multi-layer Coordination (MuC) is proposed for vehicle control. Fig. 2 shows the architecture with its components. In this architecture, inputs such as the longitudinal speed, steering angle of the front wheels and the system states (lateral speed V_y, yaw rate $\dot{\psi}$, roll angle ϕ and roll rate $\dot{\phi}$) are obtained from CarSim® as it were from a real car.

4.1 Supervision layer

The supervision layer aims to compute the references, as well as conditions under which the controller will act. The reference for β is zero and it is calculated based on the estimated car lateral speed, as shown in Eq. (12). It is also defined that the controller for β only performs whether it is outside the limits in Eq. (13). In the case of the

[1] This process ensures that integral terms appear in every factor, thereby aiding in noise reduction.

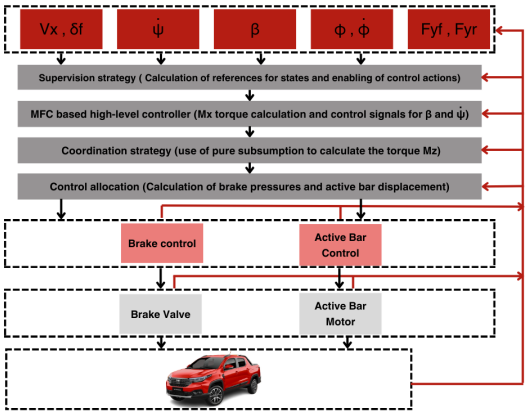


Fig. 2: UCC architecture

yaw rate, the controller's reference is calculated based in the longitudinal speed, the steering angle of the front wheels, and the coefficient of ground friction. The controller's operating range is determined by the limits described in Eq. (14). In the case of the roll angle, the reference will always be zero, as one of the controller's objectives is to reduce this parameter. The equations are:

$$\beta = \tan^{-1}(\frac{V_y}{V_x}), \tag{12}$$

$$\beta \leq \beta_{lim} = \pm 2^{\circ}, \tag{13}$$

$$\dot{\psi}_D = \min\left(\left|\frac{\mu g}{V_x}\right|, \left|G_{\dot{\psi}}\delta_f\right|\right) \cdot \text{sign}(\delta_f), \tag{14}$$

where, $G_{\dot{\psi}} = V_x/(L + K_{us}V_x^2)$, V_x is the car longitudinal speed, L is the distance between the car's axles in meters and K_{us} is the under-steer coefficient.

4.2 High level controller

The high-level controller design uses the references calculated by the supervision layer and the estimated states and calculates the torques needed to control the dynamics. In the case of lateral dynamics (which involves $\dot{\psi}$ and β), the control signal will be a torque around the car's z-axis (M_z). This signal is setting a saturation level of ±8000Nm. To control the roll angle, torque is calculated around the x-axis (M_x). This signal is limited by saturation levels of ±1500Nm. Fig. 3 shows the implementation of this controller in Matlab/Simulink®.

4.3 Coordination layer

As can be seen in Fig. 3, in the coordination phase, a Pure Subsumption strategy is used to weigh up which of the control signals corresponding to β and $\dot{\psi}$ will have more weight in the final torque calculation. This layer is not involved in controlling the roll angle.

4.4 Control allocation

At this step, the control signals calculated in the previous steps are transformed into "understandable" variables by the subsystems that act directly on the car's chassis. In the case of the torque on the z-axis, it will be transformed into individual brake pressure signals for each wheel by the ABS. In the case of M_x, it will be transformed into suspension forces relating to the active stabilizer bar (AARB) system. This step is carried out directly by the CarSim® software.

4.5 Individual actuators

The individual brake pressure signals and the motor torque signal of the AARB system from the previous step are taken in this step as references for the controllers of the car's ABS and the active stabilizer bar system, respectively. The control of these systems finally makes it possible to establish the desired behavior of the controlled dynamics. Like the previous one, this stage is implemented by CarSim® software.

5 Control tuning

In the following subsections, a demonstration of how to adjust the parameters α, K_P, and K_D of the iPD proposed controllers is performed (in this paper, we consider the integrative factor equals to zero). It should be noted that this tuning is empirically done and that the dynamics to be controlled do not require high precision in tracking the corresponding references for the system to exhibit proper behavior.

5.1 α adjustment

This parameter is set individually for each control law, and has direct influence on the system outputs (M_z and M_x torques). Since the M_z signal is composed of the sum of individual signals corresponding to the side-slip angle control and the yaw

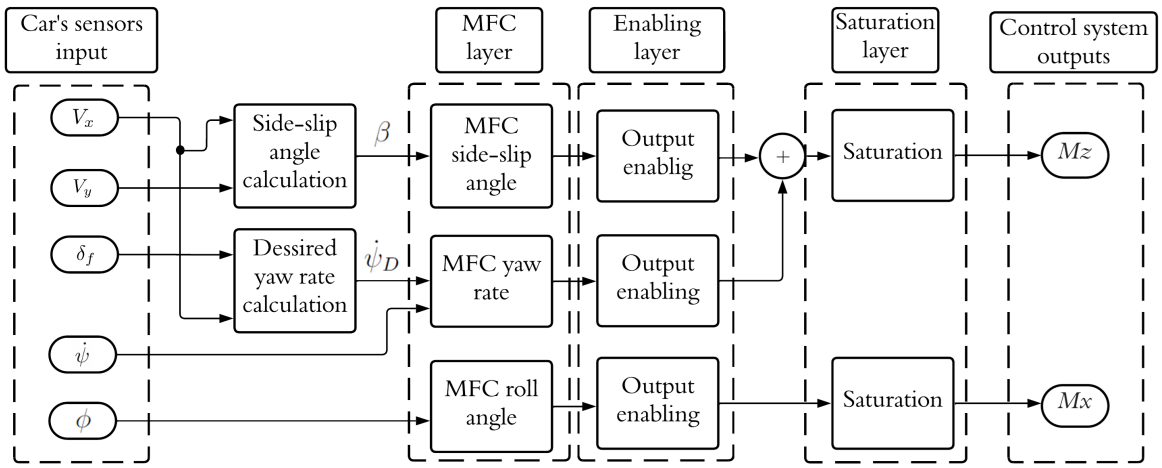


Fig. 3: UCC implementation

rate control, it is necessary to adjust simultaneously α_β and $\alpha_{\dot{\psi}}$. A bad choice of these parameters can result in an oscillatory control signal with a high saturation time and/or an unsatisfactory maneuver performance. In this work, the values of α_s are empirically obtained looking for a relationship between them and the time response of the controlled parameters. Higher α value can induce some delay in the system response while a lower α generates a more oscillatory control signal. The objective then is to find a equilibrium α. The selected α values are $\alpha_\beta = 0.01$, $\alpha_{\dot{\psi}} = 10^{-3}$ and $\alpha_\phi = 10^{-2}$.

5.2 iPD adjustment

It was verified by simulations that the variation of the K_d parameter does not have influence on the system response. The K_p was adjusted firstly for the yaw rate and side-slip angle controllers and then for the roll control. The adjustment must keep a balance between reducing the steady state error and producing an oscillatory control signal. It should be noted that in the event of a sudden maneuver (as in the case of SWD) the control must prioritize the safety over the comfort. This is evidenced by the increase in the roll rate signal. Finally, the iPD adjustment is as follows: $Kp_\beta = 20$; $Kd_\beta = 3$; $Kp_{\dot{\psi}} = 50$; $Kd_{\dot{\psi}} = 3$; $Kp_\phi = 200$; $Kd_\phi = 3$.

6 Experiment design

To validate the designed UCC, it is simulated in Matlab/Simulink® acting on the Carsim® SUV model adapted to the parameters referred in Arronte et al. (2023) in a Sine With Dwell (SWD) maneuver. This simulation will be compared to the response of the traditional PID controllers designed in Arronte et al. (2023) acting under the same conditions. To show the system's robustness, tests are performed three different μ values (Ahrenhold et al., 2023): $\mu = 0.7$ (Dry Asphalt Road), which simulates good grip and stability for vehicles, $\mu = 0.5$ (Wet Road), which corresponds to reduced traction due to a lubricating film of water, increasing the risk of skidding, and $\mu = 0.3$ (Icy Road), which simulates extremely low traction, and requires careful driving and often needs specialized tires or chains for safety.

The SWD maneuver involves accelerating the vehicle to a speed of just over 80 km/h without steering or braking, leaving it in the highest gear up to 80 km/h, and then using a robot to apply a steering control in the form of a sine wave with a dwell (momentary pause in the maximum steering angle). The purpose of the test is to evaluate the vehicle's response to a sudden evasive maneuver, such as swerving to avoid an obstacle, and to ensure compliance with regulations such as USA FMVSS 126 and UN/ECE Regulation No. 13-H Mechanical Simulation Corporation (2012). The system simulation is performed by using CarSim® and Matlab® together. Figure 4

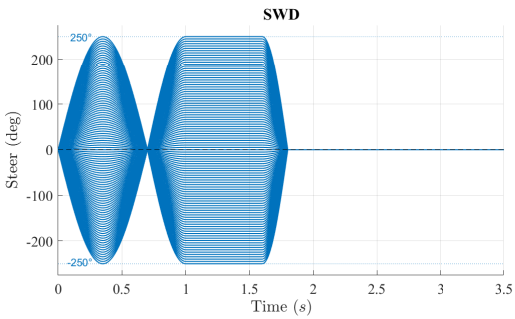


Fig. 4: Steering Angle for the Sine With Dwell maneuver

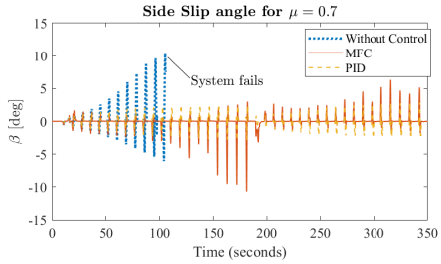
shows the waveform of the steering signal in this maneuver.

7 Simulation results

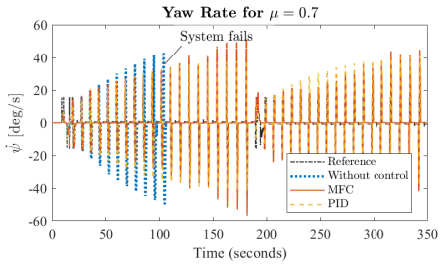
The UCC is tested by simulating a SWD maneuver as specified above for three different values of friction coefficient as described. The tests are performed for open-loop situation (without controller), PID controllers as designed in Arronte et al. (2023), and iPD MFC controllers. It is shown that for the first and second cases, stability is not even guaranteed.

7.1 Simulation for $\mu = 0.7$

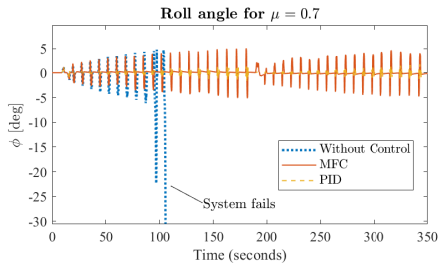
Figures 5a and 5b show the side slip angle and yaw rate in the SWD maneuver for $\mu = 0.7$. Here, both the controllers, that is PID and iPD (MFC), manage to decrease the side slip angle during the maneuver. Also, the measured yaw rate follows the reference for both controllers. It can be observed that the vehicle fails to complete the maneuver without the action of any controller (open-loop). Figures 5c and 5d show the roll angle and roll rate behavior during the maneuver. Both controllers manage to reduce these variables during the maneuver, thus demonstrating their effectiveness in controlling the rollover dynamics.



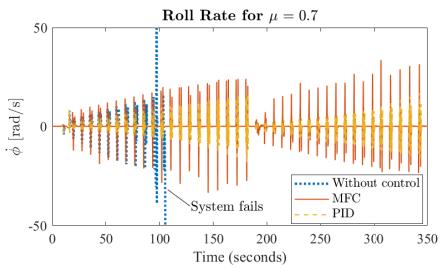
(a) Side-slip angle



(b) Yaw rate



(c) Roll angle

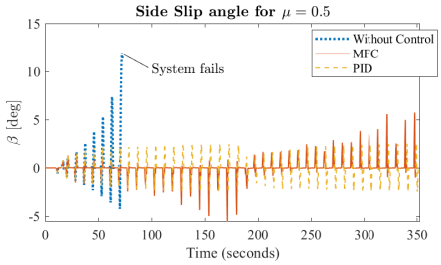


(d) Roll rate

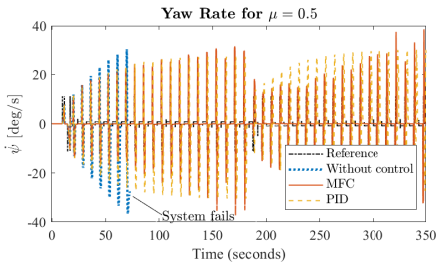
Fig. 5: Results for $\mu = 0.7$

7.2 Simulation for $\mu = 0.5$

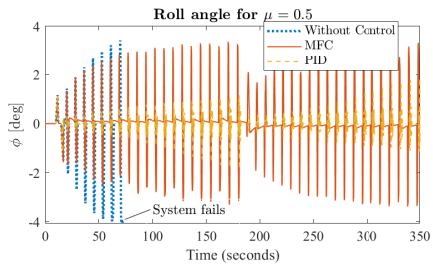
Figures 6a and 6b show the side slip angle and yaw rate respectively in the SWD maneuver with $\mu = 0.5$. Both controllers (PID and iPD MFC) effectively reduce the side-slip angle and ensure that the yaw rate follows its reference. As in the previous simulation, it can be observed that the car does not manage to complete the maneuver in open-loop (sudden stops in the curves). Figures 6c and 6d show the behavior of the roll angle and roll rate respectively. Here it is also evident the positive influence of the designed controllers in the reduction of the key parameters in the car rollover dynamics.



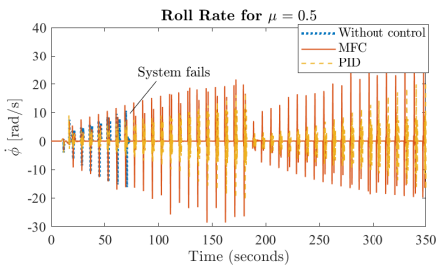
(a) Side-slip angle



(b) Yaw rate

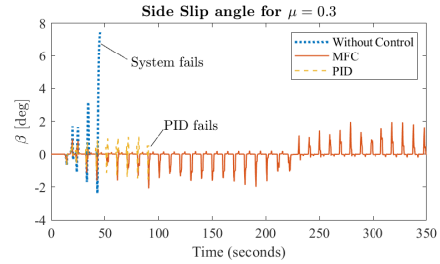


(c) Roll angle

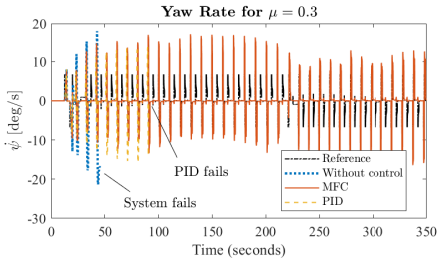


(d) Roll rate

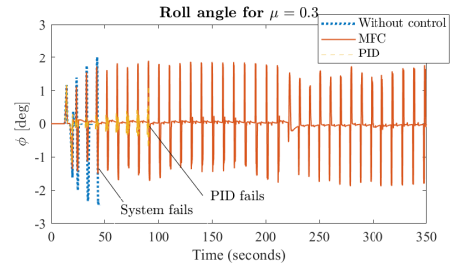
Fig. 6: Results for $\mu = 0.5$



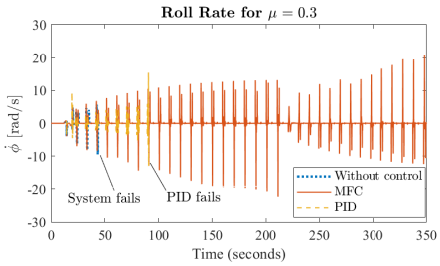
(a) Side-slip angle



(b) Yaw rate



(c) Roll angle



(d) Roll rate

Fig. 7: Results for $\mu = 0.3$

7.3 Simulation for $\mu = 0.3$

Figures 7a and 7b show the side slip angle and yaw rate respectively during the SWD maneuver with a friction coefficient of $\mu = 0.3$. It is observed that only the iPD MFC controllers are able to complete the maneuver and reduce the side slip angle and yaw rate error. Both the open-loop and the simple PID controllers fail shortly after the start of the maneuver. Figures 7c and 7d show the roll angle and roll rate and the failure can also be seen. On the other hand, it is remarkable how the designed MFC controller manages not only to complete the maneuver but also to maintain the roll dynamics in a desired range of values.

7.4 Statistical results across different friction conditions

Table 1 presents a statistical analysis of the vehicle's dynamic response under a Sine With Dwell (SWD) maneuver, considering the same three different road friction coefficients: $\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$. The results are reported for the two control strategies (PID and iPD MFC) as well as open-loop case. The cases in which stability is not guaranteed are presented in **bold numbers**.

Side-slip angle: Both controllers demonstrate a clear ability to reduce the side-slip angle. The MFC consistently achieves the lowest RMS and RMSE values, particularly at $\mu = 0.5$ and $\mu = 0.3$, where the PID controller shows slightly higher slip levels. At $\mu = 0.7$, MFC manages a lower RMS (0.627) compared to PID (0.735), and at $\mu = 0.5$, this difference becomes more significant (MFC: 0.531 vs. PID: 0.71). In the open-loop case, the side-slip angle increases substantially, peaking at 11.945° for $\mu = 0.5$, highlighting the importance of active control for maintaining lateral stability.

Yaw rate: In all conditions, the MFC achieves the lowest yaw rate RMS and RMSE values, followed by the PID controller. At $\mu = 0.7$, MFC maintains an RMS of 10.770 deg/s and a maximum yaw rate of 34.203 deg/s, while the uncontrolled vehicle reaches a peak of 49.98 deg/s. This pattern continues for $\mu = 0.5$, where MFC maintains better tracking and dampening of yaw motion.

Roll angle: Roll behavior also improves with both controllers. At $\mu = 0.7$, the uncontrolled vehicle experiences an extreme peak roll angle of 79.993, while MFC and PID limit this to 5.917° and 4.161°, respectively. At $\mu = 0.5$, the PID controller shows superior performance in minimizing the roll angle (RMS: 0.222), while MFC reaches a higher RMS (1.260) but still provides acceptable control.

Roll rate: Similar to the roll angle, the roll rate is substantially reduced by the control strategies. At $\mu = 0.7$, the uncontrolled vehicle reaches a peak roll rate of 115.104 deg/s, whereas the PID and MFC controllers keep it below 42 deg/s. The same trend is observed for $\mu = 0.5$, with the MFC showing higher values than PID at $\mu = 0.5$.

For $\mu = 0.3$, PID control and open-loop does not reach stability. Overall, the analysis demonstrates that both control strategies (PID and iPD MFC) significantly enhance vehicle stability and rollover mitigation during the SWD maneuver for $\mu = 0.7$ and $\mu = 0.5$, with the MFC generally offering superior performance in minimizing side-slip and yaw rate errors. The roll angle and roll rate are better in PID control. On the other hand, iPD MFC has a better robustness of stability, as it does not fail in the case of $\mu = 0.3$. The uncontrolled vehicle (open-loop) is not robustly stable.

8 Conclusions

This study proposes a novel Unified Chassis Control (UCC) system based on Model-Free Controllers (MFCs) to simultaneously regulate vehicle lateral (side-slip angle and yaw rate) and vertical (roll angle) dynamics during abrupt maneuvers. By leveraging ultra-local models and algebraic estimators, the MFC-based UCC eliminates the need for detailed vehicle modeling, effectively addressing challenges posed by parametric uncertainties and the nonlinear nature of traditional model-based approaches. The controller was evaluated through a Sine With Dwell (SWD) maneuver in a co-simulation environment using MATLAB/Simulink® and CarSim®, under varying road friction conditions ($\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$). Results showed that the MFC-based UCC significantly reduced side-slip angle and yaw rate errors while maintaining good roll angle stability, ensuring comprehensive lateral and vertical control. In low-friction scenarios ($\mu = 0.3$), only the MFC showed stability.

Future research should focus on automating the tuning using optimization to enhance scalability and precision, validating the approach through real-world implementations, and comparing it with alternative model-free techniques such as neural networks or fuzzy logic controllers. Additionally, extending the UCC framework to account for longitudinal dynamics or adapting it for diverse vehicle types, including electric and heavy-duty platforms, could further expand its applicability.

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Table 1: Statistical analysis of simulations (lack of robustness of stability in **bold numbers**)

		PID			MFC			No Control		
Parameter	μ	RMS[1]	RMSE[2]	MAX[3]	RMS	RMSE	MAX	RMS	RMSE	MAX
Side-slip angle	0.7	0.735	0.735	2.889	0.627	0.627	3.404	**1.826**	**1.826**	**10.33**
	0.5	0.710	0.710	2.531	0.531	0.531	2.860	**2.074**	**2.074**	**11.945**
	0.3	**0.280**	**0.280**	**1.540**	0.382	0.382	2.077	**1.387**	**1.387**	**7.466**
Yaw rate	0.7	12.086	14.751	40.857	10.770	13.659	34.203	**12.730**	**15.165**	**49.980**
	0.5	9.898	11.561	30.451	8.288	10.200	26.169	**10.303**	**10.303**	**36.629**
	0.3	**4.265**	**4.265**	**15.655**	4.739	5.695	17.267	**6.203**	**6.203**	**21.354**
Roll angle	0.7	1.335	1.335	4.161	1.646	1.646	5.917	**4.642**	**4.642**	**79.993**
	0.5	0.222	0.222	1.852	1.260	1.260	4.029	**1.340**	**1.340**	**4.050**
	0.3	**0.164**	**0.164**	**1.166**	0.598	0.598	1.893	**0.804**	**0.804**	**2.452**
Roll rate	0.7	4.092	4.092	26.348	5.745	5.745	41.935	**9.844**	**9.844**	**115.104**
	0.5	2.488	2.488	20.689	4.665	4.665	37.391	**2.799**	**2.799**	**16.156**
	0.3	**1.181**	**1.181**	**15.443**	2.320	2.320	22.313	**1.528**	**1.528**	**9.482**

[1] Root Mean Square (related to signal).
[2] Root Mean Square Error (related to reference).
[3] Maximum absolute value.

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References

Ahmadi, B. 2021. Design a reliable model predictive control for path following with application to the autonomous vehicle and considering different vehicle models. Master's thesis, University of Waterloo.

Ahrenhold, T., J. Iatropoulos, and R. Henze. 2023. Accuracy requirements for the road friction coefficient estimation of a friction-adaptive automatic emergency steer assist (esa). *Automotive and Engine Technology 8*(3): 141–151 .

Aleksander, H. and B. Edward. 2007. Robustness of side slip estimation and control algorithms for vehicle chassis control. *20th International Technical Conference on the Enhanced Safety of Vehicles (ESV)* (07-0353) .

Arronte, C.A., G.A. Ripari, B.A. Angélico, and D. Colón 2023. Design and validation of pid controller for electronic stability control. In *2023 15th IEEE International Conference on Industry Applications (INDUSCON)*, São Bernardo do Campo, São Paulo, pp. 343–350.

Bilen, Ö. 2010. Path control algorithms for autonomous steering and braking of heavy vehicles. *Göteborg: Chalmers University of Technology*: 10–52 .

Chen, X., M. Wang, and W. Wang. 2021. Unified chassis control of electric vehicles considering wheel vertical vibrations. *Sensors 21*(11): 321–342 .

Dong, H.r., S.t. Jin, and Z.s. Hou 2014. Model free adaptive control for automatic car parking systems. In *Proceeding of the 11th World Congress on Intelligent Control and Automation*, pp. 1769–1774.

Fahimi, F. 2013. Full drive-by-wire dynamic control for four-wheel-steer all-wheel-drive vehicles. *Vehicle System Dynamics 51*(3): 360–376 .

Falcone, P., F. Borrelli, J. Asgari, H.E. Tseng, and D. Hrovat. 2007. Predictive active steering control for autonomous vehicle systems. *IEEE Transactions on Control Systems Technology 15*(3): 566–580. https://doi.org/10.1109/TCST.2007.894653 .

Fliess, M. 2009. Model-free control and intelligent pid controllers: towards a possible trivialization of nonlinear control? *IFAC Proceedings Volumes 42*(10): 1531–1550 .

Fliess, M. and C. Join. 2008. Commande sans modèle et commande à modèle restreint. *e-STA Sciences et Technologies de l'Automatique 5*(4): 1–23 .

Friedman, D., G. Mattos, and J. Paver 2010. Characterizing the injury potential of a real world rollover. In *ICRASH Conference, Washington, DC*.

Gidlewski, M.J., K. JANKOWSKI, A. MUSZYŃSKI, and D. ŻARDECKI 2017, mar. Vehicle lane change automation with active steering - theoretical studies and numerical investigations. In *WCX™ 17: SAE World Congress Experience*. SAE International.

Guo, J., K. Li, J. Fan, Y. Luo, and J. Wang. 2021. Neural-fuzzy-based adaptive sliding mode automatic steering control of vision-based unmanned electric vehicles. *Chinese Journal of Mechanical Engineering 34*(1): 88. https://doi.org/10.1186/s10033-021-00597-w .

Guo, N., B. Lenzo, X. Zhang, Y. Zou, R. Zhai, and T. Zhang. 2020. A real-time nonlinear model predictive controller for yaw motion optimization of distributed drive electric vehicles. *IEEE Transactions on Vehicular Technology* 69: 4935–4946. https://doi.org/10.1109/tvt.2020.2980169 .

Lai, F. and C.Y. Jiang. 2021. Establishment and simulation analysis of 18 dof unified dynamics model of automobile chassis. *International Journal of Control, Automation and Systems 19*(7): 2323–2342. https://doi.org/10.1007/s12555-019-0750-9 .

Levine, S., C. Finn, T. Darrell, and P. Abbeel. 2016. End-to-end training of deep visuomotor policies. *Journal of Machine Learning Research 17*(39): 1–40 .

Lew, T., M. Greiff, F. Djeumou, M. Suminaka, M. Thompson, and J.K. Subosits. 2024. Risk-averse model predictive control for racing in adverse conditions. *2025 IEEE International Conference on Robotics and Automation (ICRA)*: 8508–8514. https://doi.org/10.1109/icra55743.2025.11127729 .

Li, Z., H. Chen, H. Liu, P. Wang, and X. Gong. 2022. Integrated longitudinal and lateral vehicle stability control for extreme conditions with safety dynamic requirements analysis. *IEEE Transactions on Intelligent Transportation Systems 23*(10): 19285–19298. https://doi.org/10.1109/TITS.2022.3152485 .

Mazzilli, V., S. De Pinto, L. Pascali, M. Contrino, F. Bottiglione, G. Mantriota, P. Gruber, and A. Sorniotti. 2021. Integrated chassis control: Classification, analysis and future trends. *Annual Reviews in Control* 51: 172–205. https://doi.org/10.1016/j.arcontrol.2021.01.005 .

Mechanical Simulation Corporation 2012, October. Sine with dwell test in carsim. Technical memo, Mechanical Simulation Corporation, 755 Phoenix Drive, Ann Arbor MI, 48108, USA.

Mechanical Simulation Corporation 2021, June. *Introduction to CarSim*. 755 Phoenix Drive, Ann Arbor MI, 48108, USA. Technical overview of CarSim features and capabilities.

Menhour, L., B. d'Andréa Novel, M. Fliess, D. Gruyer, and H. Mounier. 2018. An efficient model-free setting for longitudinal and lateral vehicle control: Validation through the interconnected pro-sivic/rtmaps prototyping platform. *IEEE Transactions on Intelligent Transportation Systems 19*(2): 461–475 .

Mustafa, G.I., H. Wang, and Y. Tian. 2019. Model-free adaptive fuzzy logic control for a half-car active suspension system. *Studies in Informatics and Control 28*(1): 13–24 .

Naciones Unidas (Noticias ONU). 2022, June 30. Accidentes viales: "una epidemia silenciosa y ambulante" que mata a 1,3 millones de personas por año. https://news.un.org/es/story/2022/06/1511112. Accessed 2025-06-24.

Nguyen, T.A. 2023, April. Assessing the dependence of vehicle rollover on many factors based

on new 4d graphs. *PLoS One 18*(4): e0284018 .

Owen, E.D. 2018. The benefits of four-wheel drive for a high-performance fsae electric racecar. *Cambridge: Massachusetts Institute of Technology*: 21–64 .

Pereira das Neves, G. and B. Augusto Angélico. 2022. Model-free control of mechatronic systems based on algebraic estimation. *Asian Journal of Control 24*(4): 1575–1584 .

Radac, M.B. and R.E. Precup. 2018. Data-driven model-free slip control of anti-lock braking systems using reinforcement q-learning. *Neurocomputing* 275: 317–329 .

Rajamani, R. 2012. *Vehicle Dynamics and Control* (2 ed.). New York: Springer.

Rizk, H., A. Chaibet, and A. Kribèche. 2023. Model-based control and model-free control techniques for autonomous vehicles: A technical survey. *Applied Sciences 13*(11) .

Sánchez-Morales, A., J.E. Naranjo, and M.Á. Sotelo. 2023. Lateral control for autonomous vehicles: A comparative evaluation. *Annual Reviews in Control* 56: 100926. https://doi.org/10.1016/j.arcontrol.2023.100926 .

Stolle, C.S., J.C. Bohlken, K.A. Lechtenberg, and D. Sicking 2011. Side-impact and non-tracking crash conditions.". In *90th Annual Meeting of the Transportation Research Board, Washington, DC*.

Swain, S.K., J.J. Rath, and K.C. Veluvolu. 2021. Neural network based robust lateral control for an autonomous vehicle. *Electronics 10*(4). https://doi.org/10.3390/electronics10040510 .

Wanki, C., C. Jaewoong, K. Chongkap, C. Seibum, and Y. Kyongsu. 2012. Unified chassis control for the improvement of agility, maneuverability, and lateral stability. *IEEE Transactions on Vehicular Technology 61*(3): 1008–1020. https://doi.org/10.1109/TVT.2012.2183152 .

Xia, Y., M. Lin, J. Zhang, M. Fu, C. Li, S. Li, and Y. Yang. 2021. Trajectory planning and tracking for four-wheel steering vehicle based on differential flatness and active disturbance rejection controller. *International Journal of Adaptive Control and Signal Processing 35*(11): 2214–2244 .

Yakub, F., A. Abu, S. Sarip, and Y. Mori. 2016, 01. Study of model predictive control for path-following autonomous ground vehicle control under crosswind effect. *Journal of Control Science and Engineering* 2016: 1–18 .

Zhan, W. 2017. A new methodology for reducing yaw rate estimation error. *World Journal of Engineering and Technology* 5: 12–20 .